



University of Information Technology
Department of Science and Engineering Information

THE LOOK E-COMMERCE

TRAFFIC, USERS FUNNEL & BEHAVIORS IN
WEBSITE

GROUP 6 | 24 DECEMBER, 2025



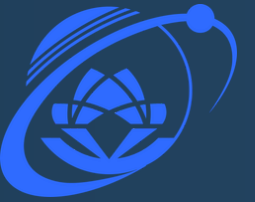


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1. INTRODUCTION



DATASET

- **THELOOK E-COMMERCE**
- **Overview:** The contents of this dataset are synthetic, and are provided to industry practitioners for the purpose of product discovery, testing, and evaluation.
- **Source:** Google Cloud



GOALS

- **Identify High-Value Traffic:** Evaluate traffic sources to prioritize long-term marketing focus.
- **Streamline User Funnel:** Detect and fix bottlenecks in the journey from search to purchase.
- **Mitigate Cart Abandonment:** Analyze drop-off factors to recover lost revenue opportunities.

TOOLS



2. DATASET OVERVIEW



Table “Users”: User Information_This table stores demographic and behavioral registration data

Column	Meaning
id	Unique ID for each user
first_name, last_name	User’s full name
email	Email address
age, gender	Age and gender
state, city, country	User’s address details
postal_code, street_address	Postal and street address
latitude, longitude	Geolocation coordinates
traffic_source	Source where the user came from (Google, Facebook, Direct, etc.)
created_at	Timestamp when the user account was created

Table “Orders”: Tracks overall order data.

Column	Meaning
order_id	Unique order identifier
user_id	User who placed the order
status	Order status
gender	Gender of the user (can be used for demographics)
created_at, shipped_at, delivered_at, returned_at	Order lifecycle timestamps
num_of_item	Total number of items in the order

2. DATASET OVERVIEW



Table “Order Items”: Each row represents a single item in an order

Column	Meaning
id	Unique ID for each order item
order_id, user_id, product_id	Foreign keys linking to the order, user, and product
inventory_item_id	Links to inventory details
status	Current item status (e.g., shipped, pending)
sale_price	Actual sale price (after discount)
created_at, shipped_at, delivered_at, returned_at	Item-specific timestamps

Table “Events”: Website Behavior - Tracks user interactions on the website

Column	Meaning
id	Unique event ID
user_id	User who triggered the event
session_id	Session identifier
created_at	Timestamp of the event
sequence_number	Order of the event in a session
event_type	Type of event (e.g home, cart, cancel,etc)
browser, ip_address	Browser used and IP address
city, state, postal_code	Geolocation details
traffic_source	Where the traffic came from
uri	Page or URL associated with the event

2. DATASET OVERVIEW



Table Name	Column	Data type	Sample value	Data Problems	Preprocessing solutions (if any)
Users	City	object (Nvarchar)	Rio Branco/ city	Contains null values and value 'city'	Replace with 'Unknown'
	created at	object (Nvarchar)	2022-09-02 04:03:00 UTC	Unclear date format	Split and Convert to datetime format
	Age	object (Nvarchar)	22	Not grouping age	Create age segmentation columns
	First name+ Last name	object (Nvarchar)	Priscilla/ Keith	2 different columns→ Difficulty in identifying top customers when visualizing charts	Create a new column that combines the first name and last name.
Events	city	object (Nvarchar)	Rio Branco/ city	Contains null values and value 'city'	Replace null with 'unknown'
	created at	object (Nvarchar)	2022-09-02 04:03:00 UTC	Unclear date format	Split and Convert to datetime format
Order Items	created at	object (Nvarchar)	2022-09-02 04:03:00 UTC	Unclear date format Contain null values (shipped, delivered, returned)	Split and Convert to datetime format Create flag columns
	shipped at				
	delivered at				
	returned at				
Orders	shipped at	object (Nvarchar)	2022-09-02 04:03:00 UTC	Unclear date format Contain null values (shipped, delivered, returned)	Split and Convert to datetime format Create flag columns
	delivered at				
	returned at				
	created at				

THE LOOK E-COMMERCE WEBSITE PERFORMANCE

Customer Registration and Visit Overview

Total Registered Users

100.0K

Total Visitors

80.0K

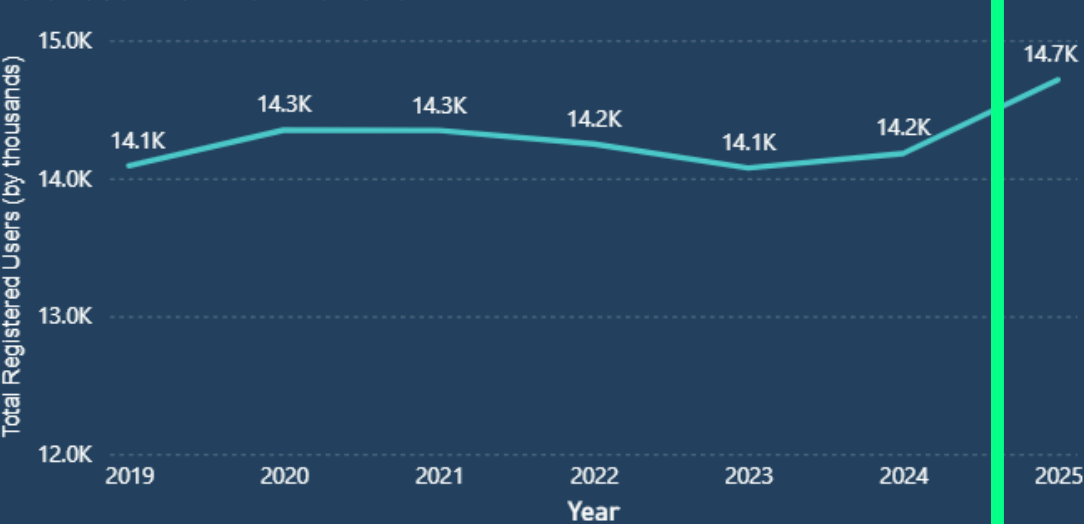
Total Session

681.7K

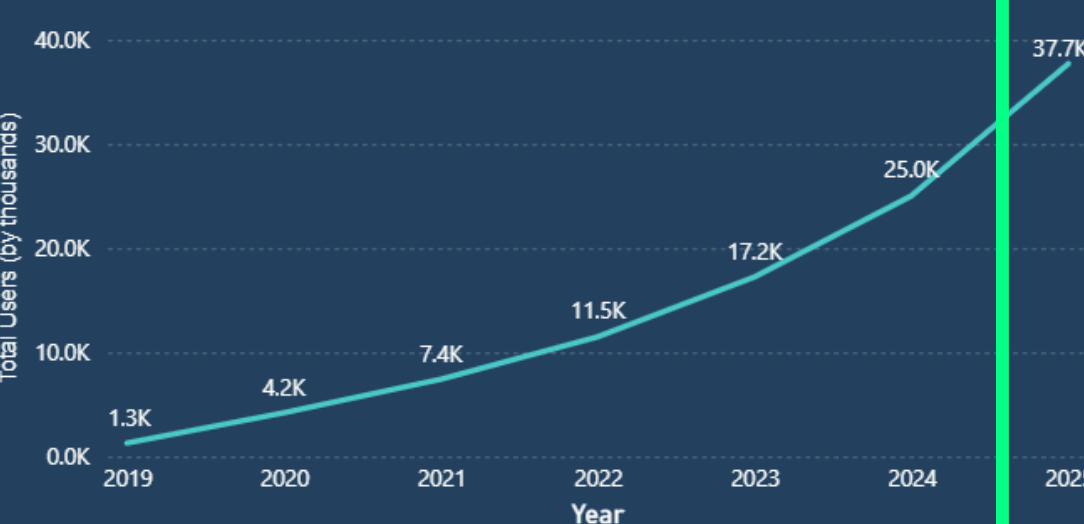
Sessions per User

8.5

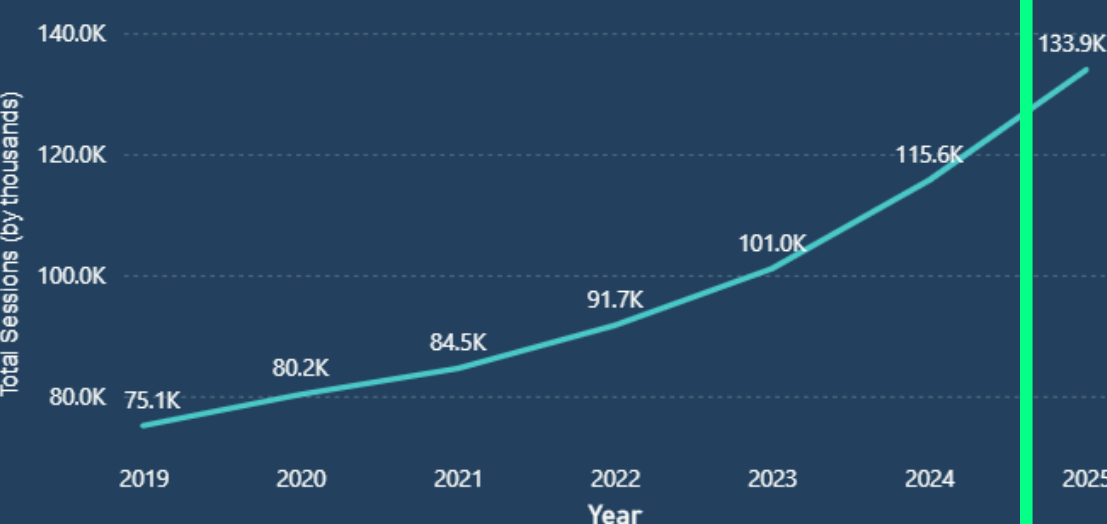
Total User from 2019 to 2025



Total Visitors from 2019 to 2025



Total Session from 2019 to 2025



- Total visitors accounted for ~80% of total registered users
- On average, users visit the website ~ 8 times.
- Overall, there will be significant growth in the number of customers and their interactions on the website from 2019-2025.
- Specially, in 2025, only covering half year, user acquisition and engagement metrics have already surpassed previous years.

THE LOOK E-COMMERCE WEBSITE PERFORMANCE



Ordering overview

Total Orders

125.3K

Total Ordered Users

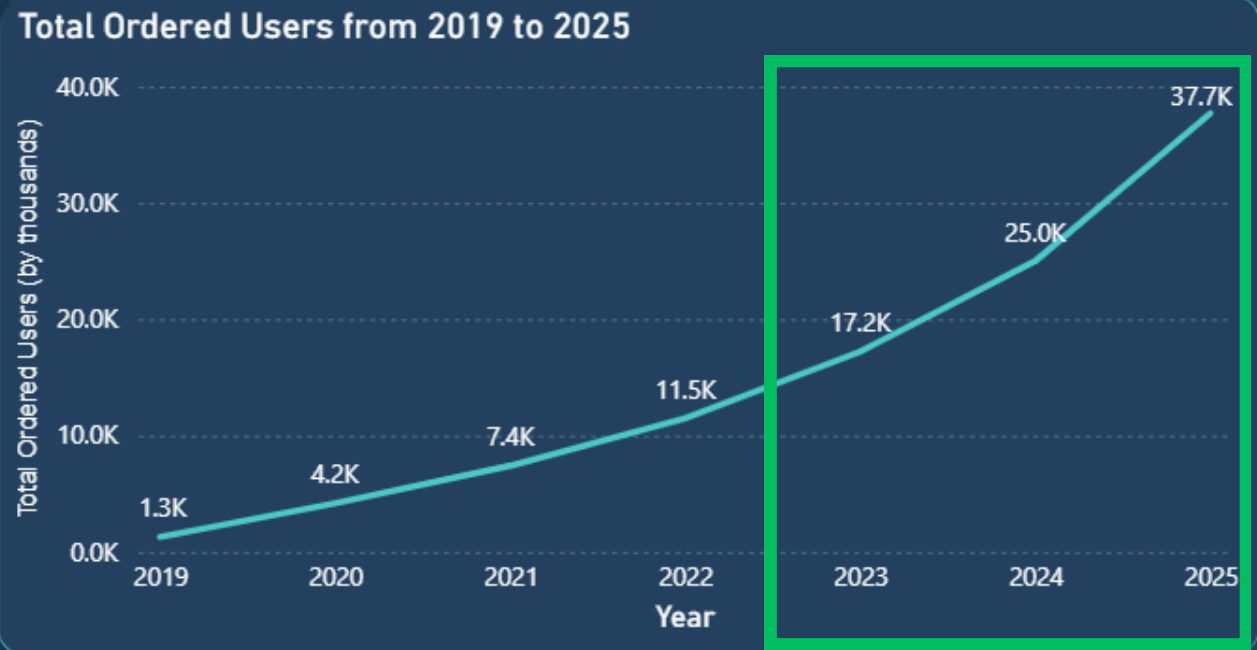
80.0K

Orders per User

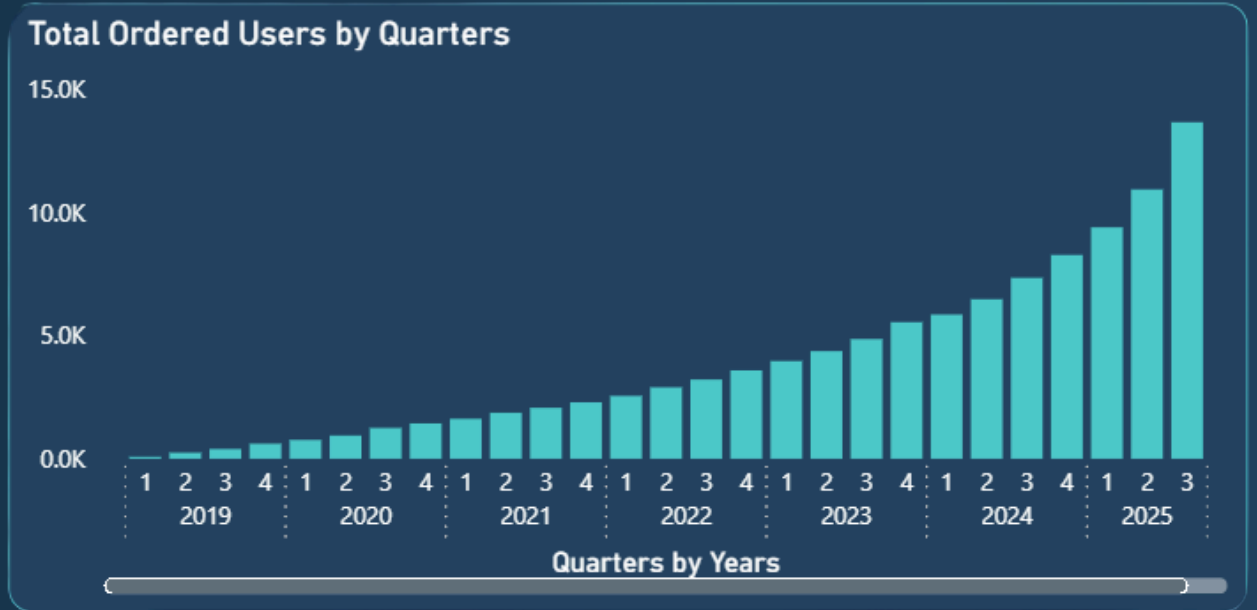
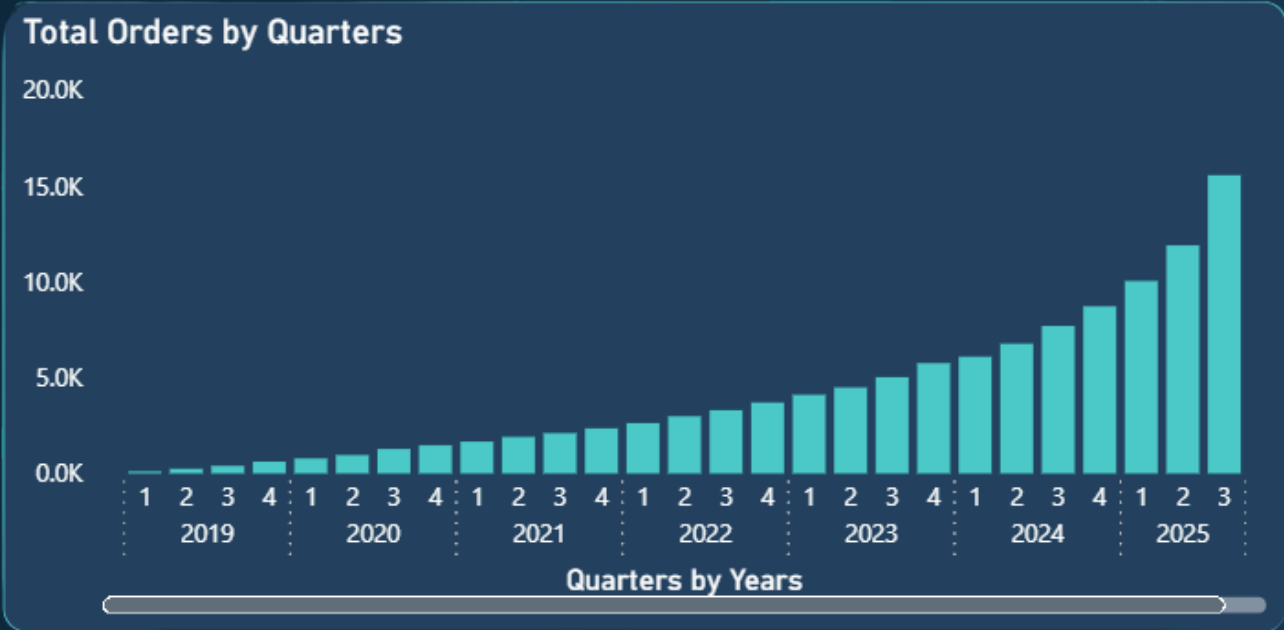
1.6

Cart Abandonment Rate (%)

57.9



The platform is experiencing exponential hyper-growth, with H1 2025 performance already nearly doubling 2024's full-year records in both order volume and active user base.



The platform demonstrates consistent quarter-over-quarter growth and peaks in the 4th quarter of year, with strong business momentum completely overriding typical post-holiday seasonal dips.

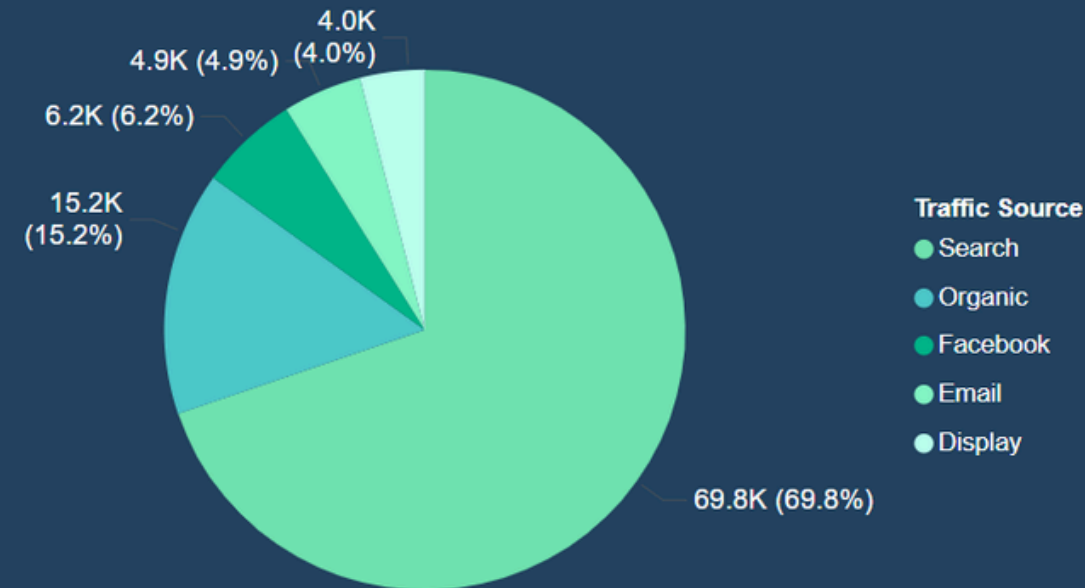
ANALYSIS OF TRAFFIC SOURCES



User and Traffic Acquisition Analysis

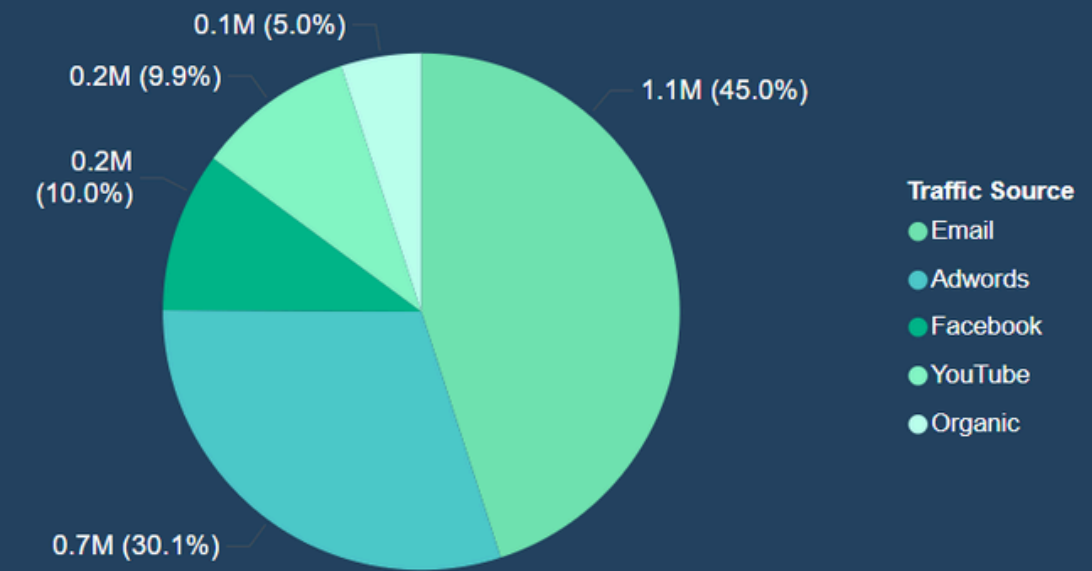
Which traffic that customers register from?

(Total register by signup traffic source)



Which traffic having an acquisition effectiveness?

(Total visitors by event traffic source)



Signup Traffic Source

- **Paid Traffic:** Search, Facebook, Display
- **Organic Traffic:** Organic, Email
- ~70% registers are from Search traffic
- Organic is accounted for 15% of total registers
- Organic traffic with the long-term goal of reaching new registers since its low cost.
- Facebook, Email, Display are both <10% of total registers

Event Traffic Source

- **Paid Traffic:** Adwords, Facebook, Youtube
- **Organic Traffic:** Organic, Email
- Total visitors are mainly from Email (45%), Adwords traffic (~30%) → working effectively in attracting users.
- Only 6.3% of total visitors from Organic
- Social media traffics are accounted for ~19% of total visitors

- Potential Traffic: Search traffic
- Traffic needs improvement: Organic traffic

- Long-term investment: Email and Adwords traffic
- Potential Traffic: Youtube and Facebook (Maintain but should not invest a high cost)
- Traffic needs improvement: Organic traffic

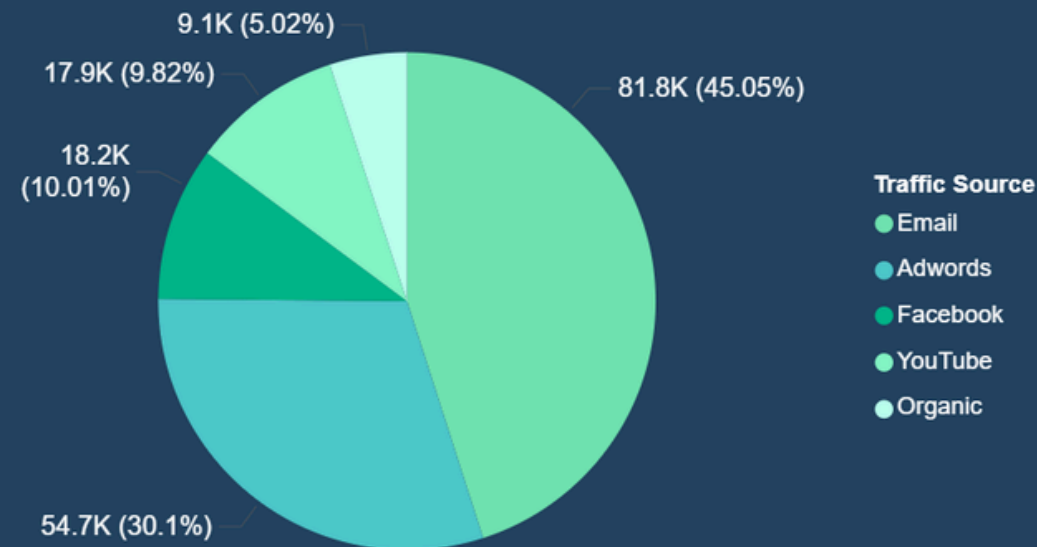


ANALYSIS OF TRAFFIC SOURCES

User and Traffic Acquisition Analysis

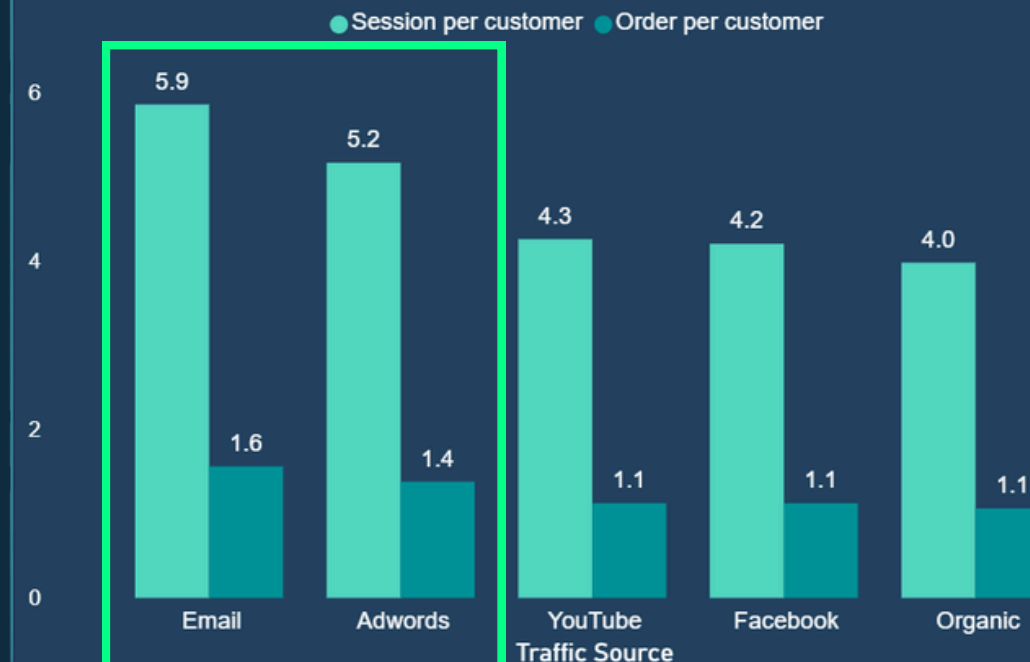
Which channel generates the most purchases?

(Total purchase by event traffic source)

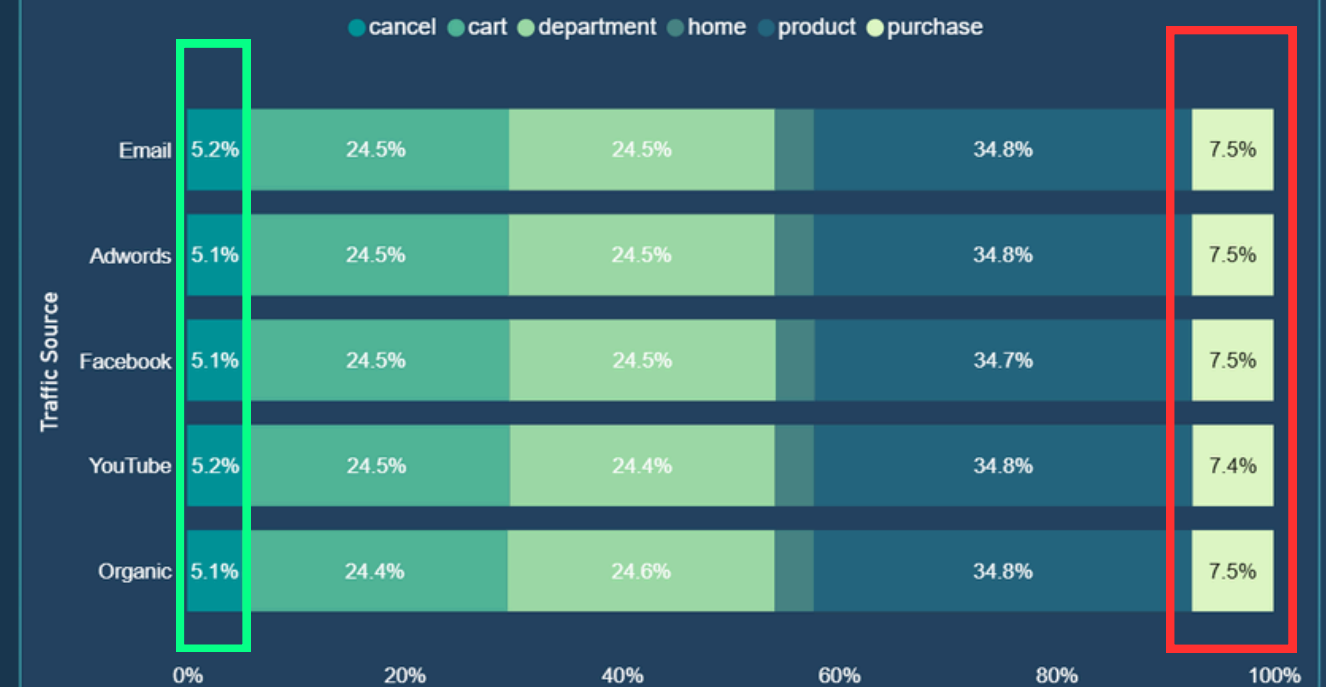


Which channels bring in the most engaged customers?

(Total purchase by event traffic source)



Total Orders (by orders status)

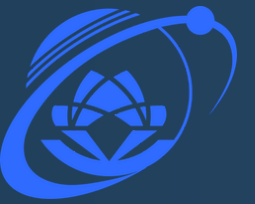


Total orders are mainly from Email (45.05%), Adwords traffic (30.1%)

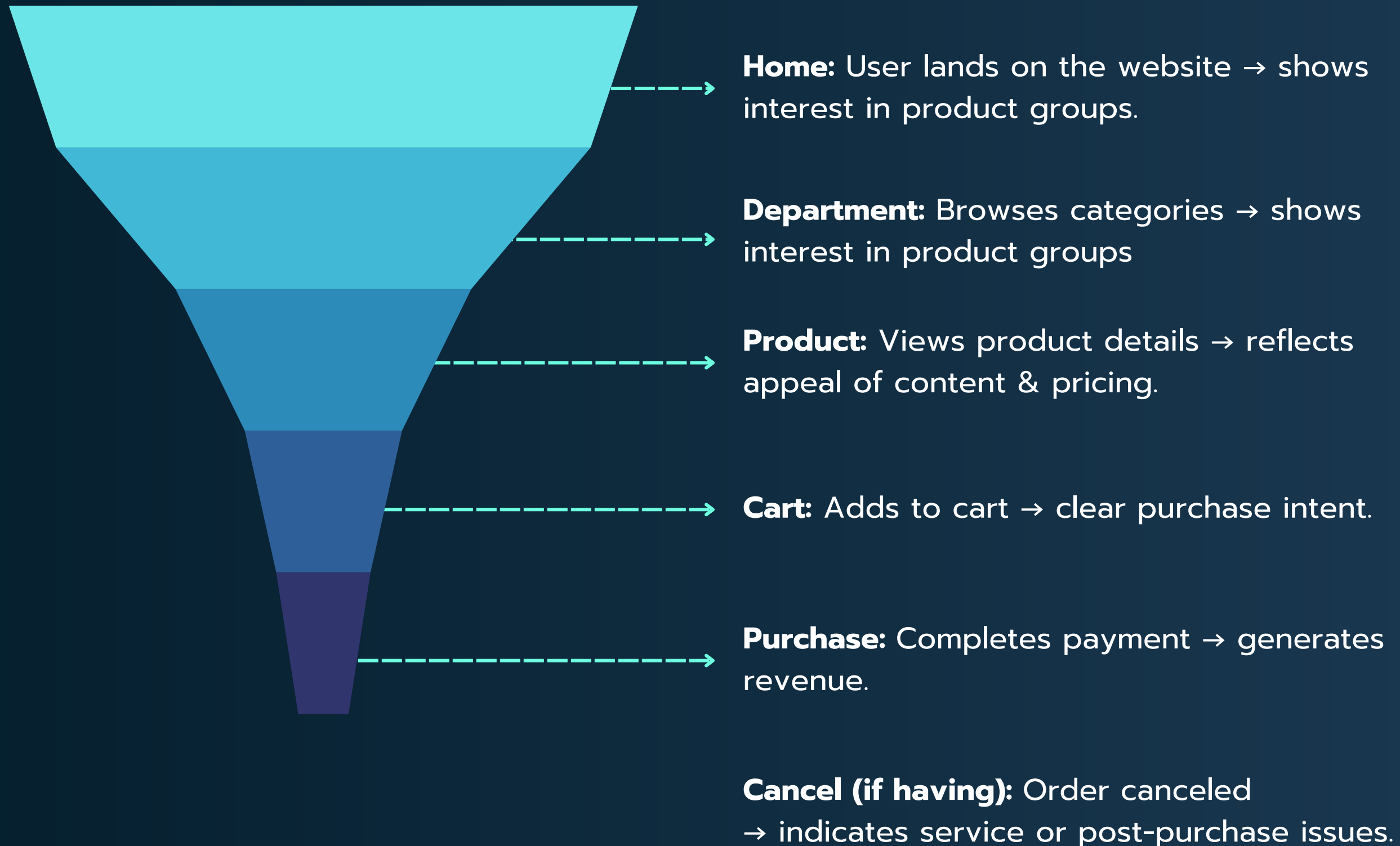
- **Higher Purchase Frequency:** Email and Adwords users show a stronger retention rate, averaging nearly 2 orders per customer compared to just ~1 from other sources.
- **Better Conversion Efficiency:** Email and Adwords users convert faster, purchasing every ~3 sessions, while Organic and Social traffic typically require 4+ sessions to buy.

- **Low Cancellation Rate:** Cancellations remain minimal (~5%) across all channels.
- **Low Conversion:** Purchase rate is low (~7.5%). Users are primarily browsing and adding items to cart rather than completing the checkout.

ANALYSIS OF USER BEHAVIORS IN PURCHASING FUNNEL



User Funnel



3 main stages:

- **Exploration:** Home → Department → Product
- **Conversion:** Cart → Purchase
- **Post-purchase:** Cancel

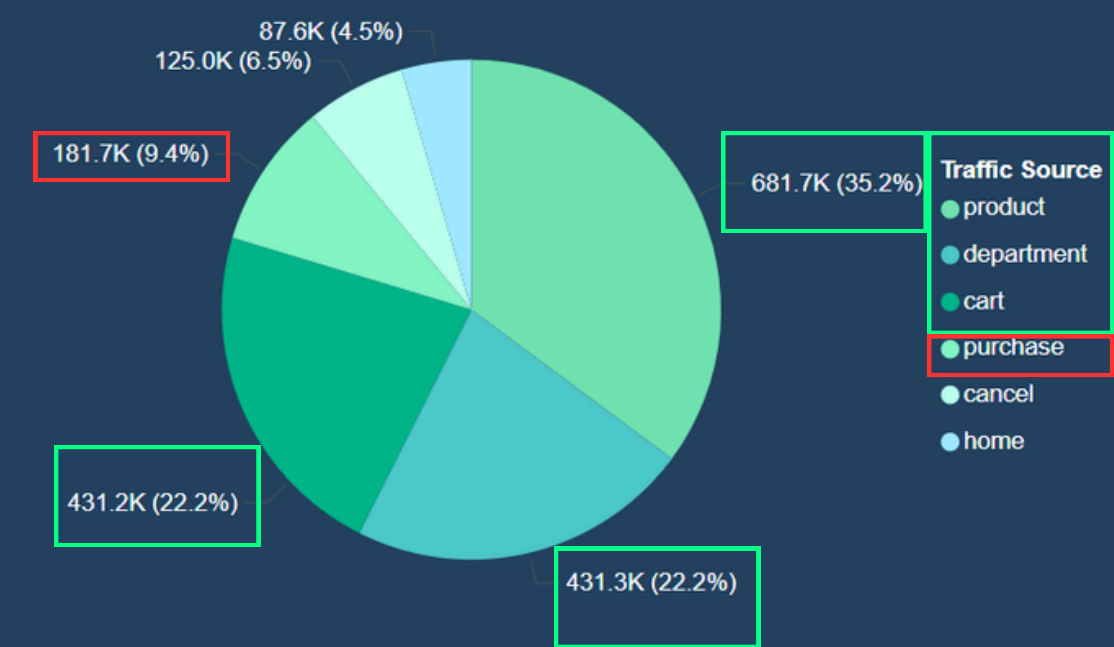
ANALYSIS OF USER BEHAVIORS IN PURCHASING FUNNEL



User Funnel

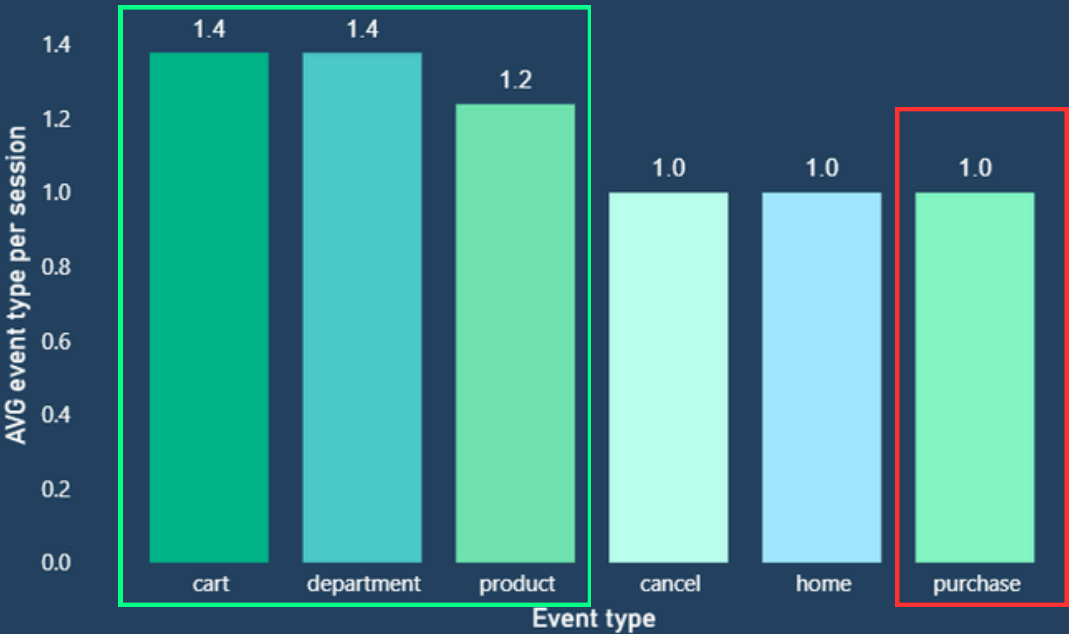
What are the most common users actions across sessions ?

(Total sessions by event types)



Which events types are likely to occur multiple times in a single session?

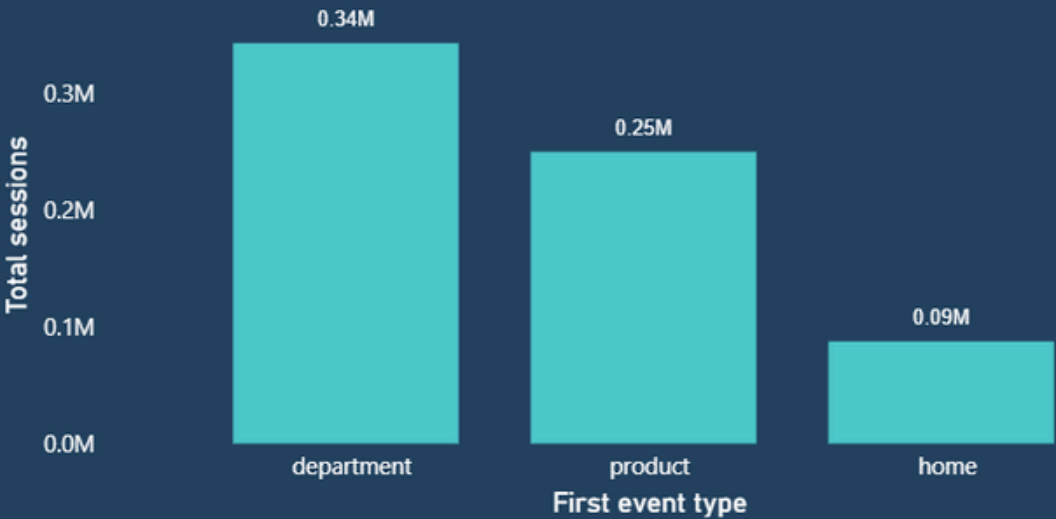
(Based on AVG event count per sessions)



- Most common user actions: Department, Cart, Product
- Purchases event are only 9.3% of total session → less occur
- In a Session, product, cart and department stages occur >1 time, and purchase step is conducted 1 time
- Most common entry point: Department
- Most common end point: Product

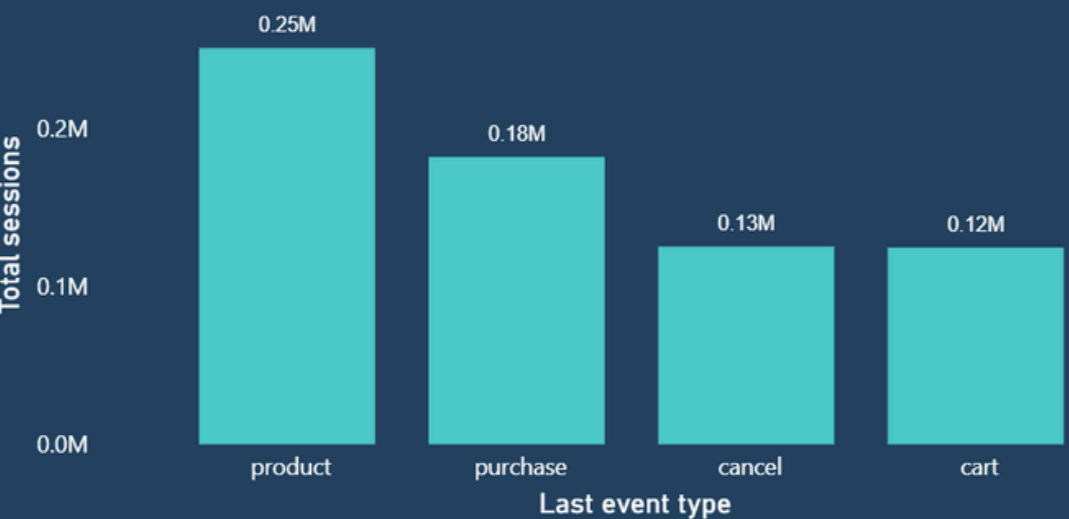
What is the most common entry point at the beginning of a session?

(Total Sessions based on First Step in the Conversion Funnel_Sequence number =1)



At which step do users drop off most frequently during their session?

(Sessions by Highest Reached Step in the Conversion Funnel)



Most users spend time exploring and products and categories but convert poorly → Tend to compare and consider carefully before buying

ANALYSIS OF USER BEHAVIORS IN PURCHASING FUNNEL

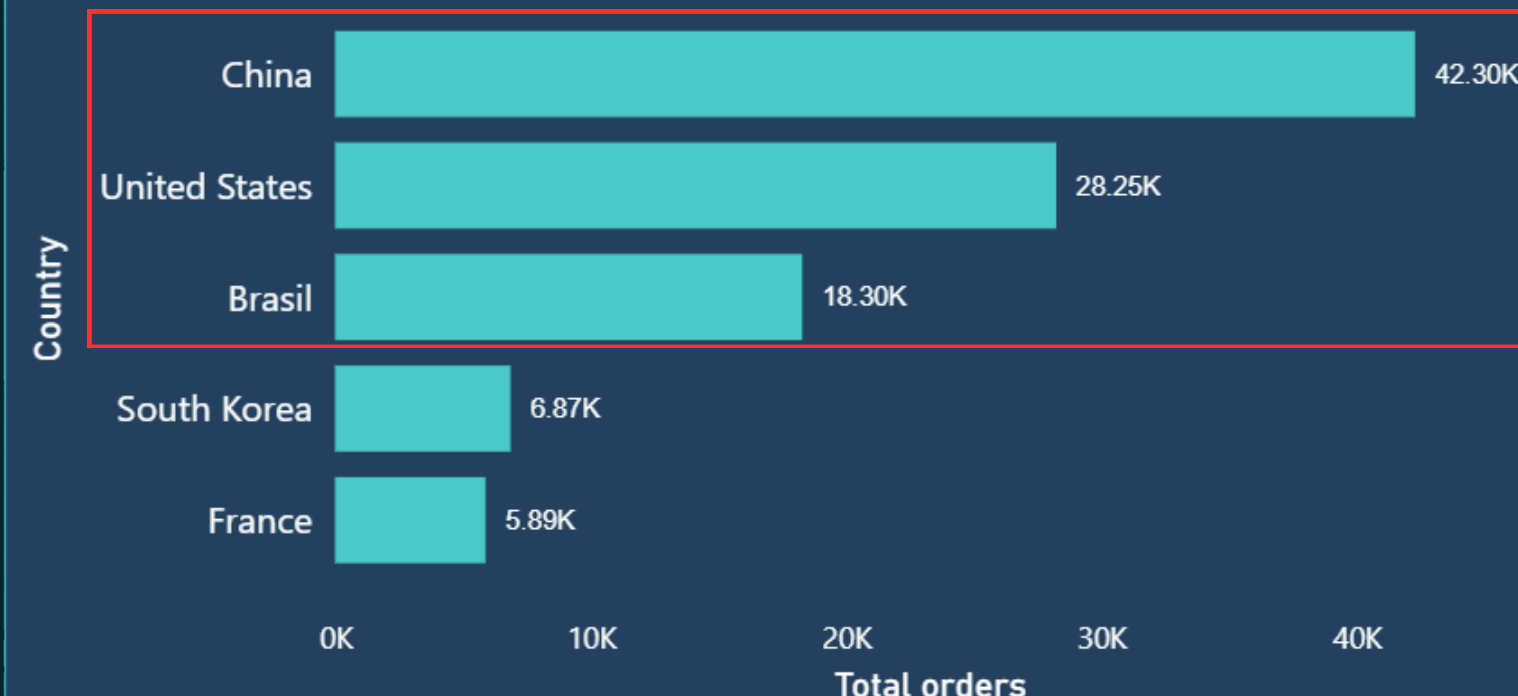


Target Customer

- **Location: 15 countries**

Top 5 countries having highest orders

(Top orders by country)



Top 5 cities having highest orders

(Top orders by city)



- **High Market Concentration:** The Top 3 countries (China, US, Brazil) dominate sales. Notably, Brazil (3rd) generates ~2.7x more orders than South Korea (4th).
- **City-Level Dominance:** China dominates the city rankings with 4 out of the top 5 spots (Shanghai, Beijing, Shenzhen, Dongguan).
- **Seoul's High Concentration:** Even though South Korea is only 4th globally, Seoul is the #3 city. This shows users there are very concentrated in the capital.

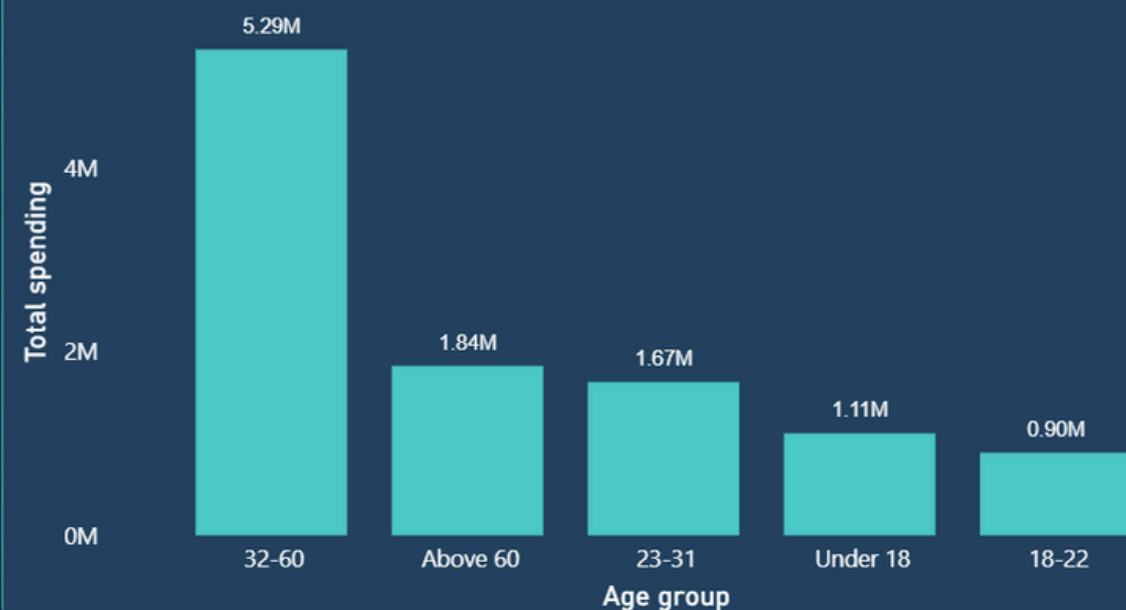
ANALYSIS OF USER BEHAVIORS IN PURCHASING FUNNEL



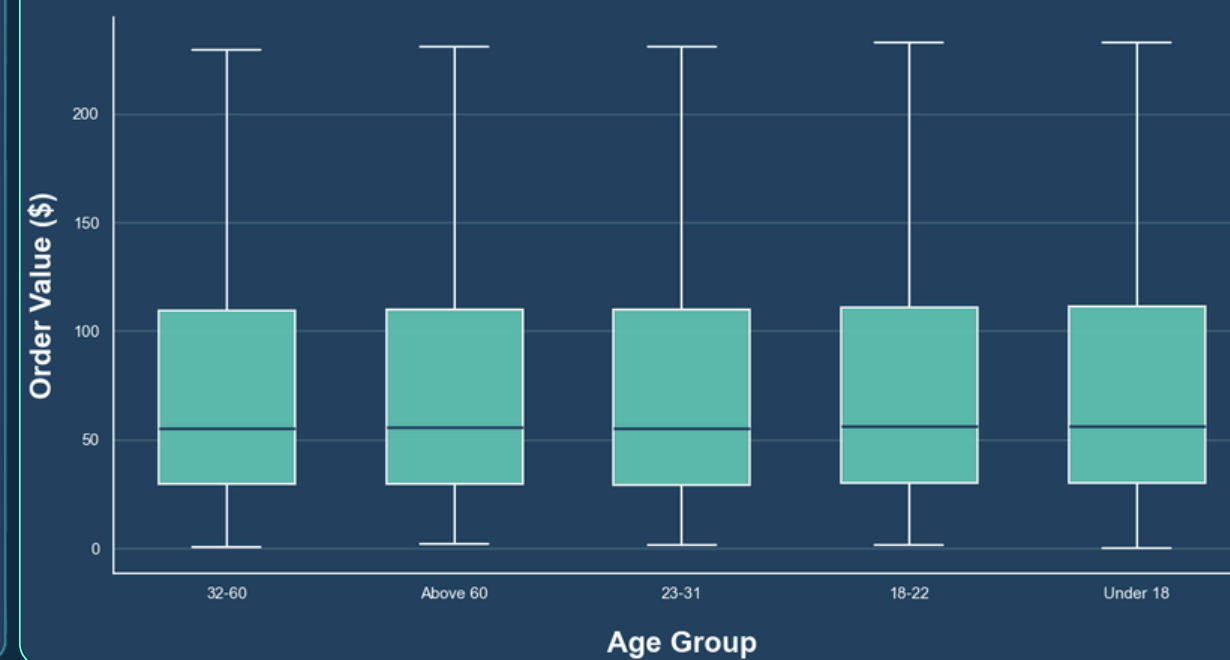
Target Customer

- Age Group

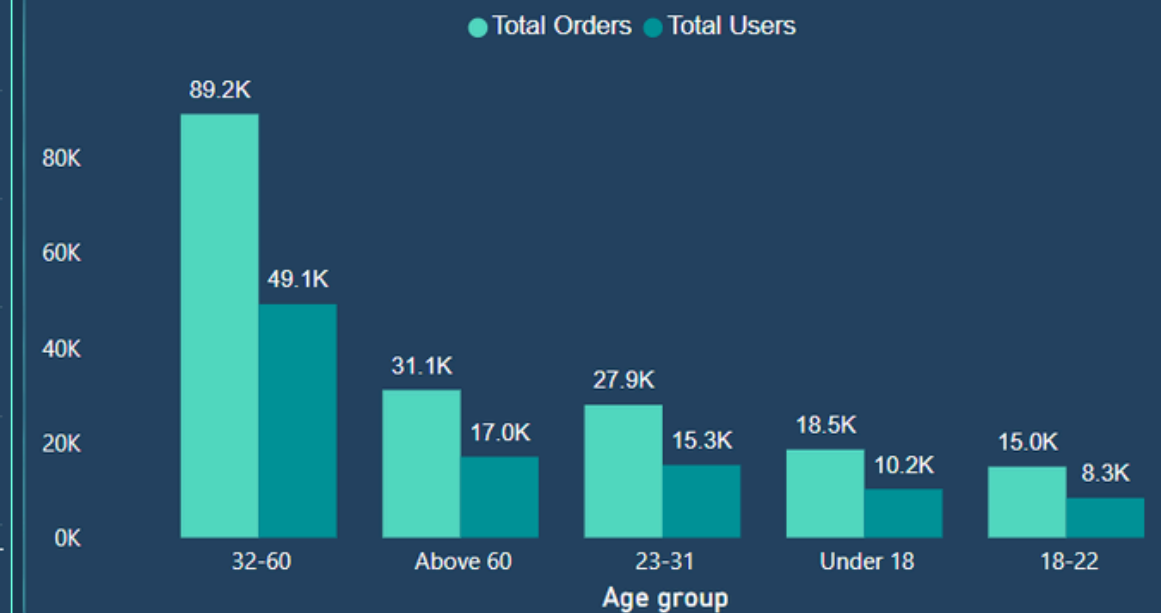
Which age group having highest spending ?



Comparing Order Value (AOV) across groups



Total Users & Orders by Age group



- **Spending Aligns with Income:** The 32-60 age group is the top spender (\$5.29M), followed by older adults (>60) and younger groups, which matches their expected earning power.
- **Similar Spending Habits:** Surprisingly, the Average Order Value (AOV) is identical across all ages, with everyone typically spending ~\$50 per order (max ~\$200).
- **Volume-Driven Dominance:** The 32-60 group leads not because they buy more expensive items, but simply because they are the largest customer base, generating the highest number of orders.

ANALYSIS OF USER BEHAVIORS IN PURCHASING FUNNEL



Target Customer

- User Segmentation

Customer Segmentation by RFM Score		
Segment	Total Customers	Total Orders
New Customers	12507	12507
Promising	11214	11214
Champions	9407	27450
Cannot Lose Them	9270	11085
Lost customers	7445	7508
Loyal	6973	16926
At Risk	6907	15928
Hibernating customers	6175	6175
About To Sleep	4727	4825
Potential Loyalist	3024	6973
Need Attention	2331	4662

Customer Segmentation by RFM Score		
Segment	Total Revenue	AOV
Promising	1,762,918.45	157.21
Cannot Lose Them	1,674,646.08	151.07
Champions	2,847,029.53	103.72
Loyal	1,548,210.30	91.47
At Risk	1,404,999.95	88.21
About To Sleep	316,553.63	65.61
Need Attention	216,558.22	46.45
Hibernating customers	211,345.52	34.23
Lost customers	253,262.63	33.73
New Customers	387,847.79	31.01
Potential Loyalist	181,824.38	26.08



- The New, Promising segments lead in volume (~23.7K), yet Lost customers (7.4K) currently outnumber Loyal ones (6.9K) → **While customer acquisition is strong, retention performance is underperforming relative to the churn rate.**
- Champions generate the highest Total Revenue (\$2.8M) with an AOV of \$103, whereas the Promising segment holds the highest AOV at \$157 → **AOV demonstrates a downward trend as customers evolve into deeper loyalty stages.**
- New Customers rank first in user count (12.5K) but last in AOV (\$31) → **Revenue from this segment is driven by transaction volume rather than value.**



KEY FINDINGS

PeakSeason

4th
Quarter

Cart Abandonment Rate

58%

Purchase Rate

7.5%

User segmentation

- **VIP Segments:** Focus on Champions (highest Revenue) and Promising (highest AOV) to secure long-term value.
- **AOV Trend:** The AOV drops from \$157 (Promising) to \$103 (Champions) and \$91 (Loyal) → demonstrates a downward trend as customers evolve into deeper loyalty stages.
- **Retention Risk:** Lost customers currently outnumber Loyal customers → signaling underperformance in retention efforts.
- **New Customers:** This group ranks first in volume but last in spending), driving transaction counts rather than revenue value.

Traffic Sources Effectiveness

- **High Performance:** Email & Adwords account for ~75% of total orders and convert faster, typically requiring only ~3 sessions to purchase.
- **Low Performance:** Organic & Social traffic (Facebook, YouTube) typically require 4+ sessions to convert. Organic traffic remains low, contributing only 6.3% of visitors.

Target Customers

- **Location:** Top markets are China, USA, and Brazil. City-level dominance is led by Shanghai, Beijing, and Seoul.
- **Age & Spending:** The 32-60 age group is the top spender due to volume. However, the AOV is consistent at ~\$50 across all age groups.

User Behavior

- **Shopping Habits:** Users spend significant time exploring and comparing products carefully, which contributes to poor conversion rates.
- **User Journey:** The most common entry point is Department and the exit point is Product. Users repeatedly view products and carts multiple times per session before buying.

The background is a dark blue gradient with several overlapping circles of varying sizes. Some circles are a lighter shade of blue, while others are a darker shade. There are also several clusters of small white dots arranged in patterns: a 2x8 grid in the top left, a vertical line of 12 dots on the right side, a vertical line of 12 dots on the left side, and a 2x8 grid in the bottom right.

THANK YOU